

	Lab	Parameters (B)	Tokens trained (B)	Ratio Tokens/Params ALScore (Churchilla scaling@20:1) "ALScore" is a quick and dirty rating of the model's power. The formula is: Sq Root of (Parameters x Tokens) + 300.	MLLM	MLLM	GPQA Training dataset	Pro	Announced	Public?	Y	Paper / Repo	Arch	Notes
AuroraGPT (ScienceGPT)	Argonne National Laboratory	2000	30000	15.1					TBA					Three models targeted in Jul2024: AuroraGPT-7B-P (Ponte Vecchio GPU testing) AuroraGPT-7B-A (Aurora) AuroraGPT-7B-A-S (Aurora + Science).
	xAI	27000							TBA				MoE	Training Jul-Dec 2024 on 100,000 H100s. Due December 2024.
	OpenAI	27000	114000						TBA				MoE	Due 2025.
	OpenAI								TBA					Due 2025.
	OpenAI								TBA					Training Oct/2024-Feb/2024 on 100,000 H100s. Due 2025.
Llama 4	Microsoft	600	40000	20.4			7.6		TBA			https://www.reuters.com/technology/microsoft-reveals-new-ai-model-completes-with-google-openai-information-report-2024-06-06/	Dense	Potential failed training run 2024. MAI=Microsoft artificial intelligence. MSFT CTO statement: https://archive.md/XRSgS
	OpenAI								TBA					Due 2025.
o4	OpenAI	5000	100000	20.1			74.5		Dec/2024			https://research.bloom.ai/	MoE	SoTA model for Dec2024. Parameter estimate is very rough centrepont for range 400B-52T.
	International	0.395	2000	5,064.1			0.1		Dec/2024			https://arxiv.org/abs/2412.13663v1	Dense	"a proper workhorse model, for retrieval, classification, etc." https://bky.app/profile/howard_fm/post/3ldo2zafps62x
ModernBERT	Granite 3.1 8B		12000	1,500.1			1.0		Dec/2024			https://github.com/ibm-granite/granite-3.1-language-models?readme=usfile	Dense	
	IBM	8	2200	245.1			0.5	60.77	17.53	4.14		Dec/2024		"trained by IBM, Princeton, CMU, and UIUC on completely open data. At inference time, the model demonstrates 2.5x throughput improvement and 2x latency speedup compared to standard transformers in vLLM."
Bamba-9B	CMU	9	2200	245.1			0.7	91.8		75.7		Dec/2024		"01-2024-12-17 sets new state-of-the-art results on several benchmarks, improving cost-efficiency and performance."
	OpenAI	200	20000	100.1			0.7	73.1	42.5	34.1		Dec/2024		"We conducted a single large-scale pretraining run on the 7B model, using 1024 H100 GPU chips, leveraging 14 trillion tokens... upscaled the 7B model to a 10B parameters model by duplicating the redundant layers and continuing pre-training with..."
o1-2024-12-17	Falcon 3	10	16000	1,600.1			0.7					Dec/2024		
	Command R7B	7	2000	286.1			0.4	28.5	7.7			Dec/2024		
Maya	Cohere	8	4800	600.1			0.7					Dec/2024		VLM.
	Meta AI	8	4500	563.1			0.6	57.4				Dec/2024		Byte Latent Transformer (BLT), a new byte-level LLM architecture that, for the first time, matches tokenization-based LLM performance
BLT	Microsoft	14	10000	715.1			1.2	84.8	70.4	56.1		Dec/2024		Available on HF from 22/Dec/2024.
	phi-4	30	30000	1,000.1			3.2	87	76.4	62.1		Dec/2024		Gemini 2.0 Flash was first model released, 11/Dec/2024. "New Modalities: Gemini 2.0 introduces native image generation and controllable text-to-speech capabilities" Announcement: https://blog.google/technology/google-deepmind/google-gemini-ai-update/
Gemini 2.0 Flash exp	Google DeepMind	7	2000	286.1			0.4	60.97				Dec/2024		"Fully Open Source" with pre-training code, configurations, training and fine-tuning datasets, and intermediate checkpoints.
	InternVL 2.5	78	18120	233.1			0.4	86.1	71.1	49		Dec/2024		o1 reasoning model copy. Benchmarks are estimates based on Qwen2.5 72B Instruct as the base LLM (InternVL 2.5=InternVT-6B-448px-V2.5 5.8B + Qwen2.5-72B-Instruct). "Notably, Qwen2-VL processed a cumulative total of 1.4T tokens, while our..."
Llama 3.3	Meta AI	70	15000	215.1			3.4	86	68.9	50.5		Dec/2024		Drop-in replacement for Llama 3.1 70B, comparable performance to Llama 3.1 405B.
	LG	32	6500	204.1			1.5	78.3		39.7		Dec/2024		"EXAONE"= "EXpert AI for EveryONE". Training tokens/ratio dropped from EXAONE-3.7 8B with 8T (Aug/2024) to this (Dec/2024) 7.8B with 9T to 32B with 6.5T.
EXAONE-3.5	DeepThought-8B	8	15000	1,875.1			1.2					Dec/2024		o1 reasoning model copy. No evals. Llama 3.1 8B base.
	Sailor2	20	18510	926.1			2.0					Dec/2024		SEA languages. Continual pretraining based on Qwen2.5. Project page: https://sea-sailor.github.io/blog/sailor2/
Pleias 1.0	Pleias	3	1086	362.1			0.2					Dec/2024		Trained on the Jean Zay supercomputer, 192x H100s for 20 days. Dataset is new CC + Synthetic: https://huggingface.co/datasets/Pleias/common_corpus
	OpenAI	200	20000	100.1			6.7	92.3	91	79		Dec/2024		"a version of our most intelligent model that thinks longer for the most reliable responses" System card about safety only: https://cdn.openai.com/o1-system-card-202412205.pdf
o1	Amazon	90	10000	112.1			3.2	85.9		46.9		Dec/2024		Multimodal, same performance as Llama 3.2 90B . . . est 90B. Model card was hidden: https://assets.amazon.science/9fa3ae416274ab2bde0911f1ebc6b830/the-amazon-nova-family-of-models-technical-report-and-model-card.pdf via https://www.amazon.science/publications/9fa3ae416274ab2bde0911f1ebc6b830
	DisTrio 15B	15	100	7.1			0.1	23.48				Dec/2024		About 14 DGXes scattered around the globe. Sometimes more sometimes less, it varies depending on availability. On average, around 112 H100s." https://x.com/bloc97/status/1863675225810043331 "we introduce DisTrio, a family of architecture..."
INTELLECT-1	Prime Intellect	10	1000	100.1			0.3	49.89		28.32		Nov/2024		Training complete 22/Nov/2024. Fully distributed training: "the first decentralized training run of a 10-billion-parameter model, inviting anyone to contribute compute and participate. This brings us one step closer towards open source AGI."
	QwQ-32B	32	18000	563.1			2.5					Nov/2024		o1 reasoning model copy. Scores 1/5 on latest ALPrompt 2024 H2. Qwen with Question=QwQ
Teuken-7B	OpenGPT-X	7	4000	572.1			0.6	50				Nov/2024		24 EU languages (60% non-English): bg, cs, da, de, el, en, es, et, fi, fr, ga, gr, hu, it, it, lt, lv, mt, nl, pl, pt, ro, sk, sl, sv. https://opengpt-x.de/models/teuken-7b-de/ & paper date is Sep/2024.
	OLMo 2	13	5600	431.1			0.9	68.6				Nov/2024		Open Language Model (OLMo) 2 Apache 2.0 license for research and educational use. Paper coming. Data: 5 trillion tokens (1.2 epochs of 4T tokens) + 100B tokens (3 runs) + 300B tokens (1 run) merged. https://huggingface.co/allenai/OLMo-2-112b
Bi-Mamba	CMU	2.7	1260	467.1			0.2					Nov/2024		Unreleased, but will be replicated. "a scalable and powerful 1-bit Mamba architecture designed for more efficient large language models"
	Moonshot AI	1000	2000	20.1			1.5					Nov/2024		o1 reasoning model copy, maths only. Very little info available. Chinese. Long context. No paper.
Marco-o1	Alibaba	7	7000	1,000.1			0.7					Nov/2024		o1 reasoning model copy. No evals. Qwen2-7B-Instruct with a combination of the filtered Open-O1 CoT dataset, Marco-o1 CoT dataset, and Marco-o1 Instruction dataset.
	TOLU 3	70	15600	223.1			3.5	93.1	65.8	45.1		Nov/2024		Llama 3.1 post-training, worse performance on most benchmarks. Post training methods include new Reinforcement Learning with Verifiable Rewards (RLVR). "We perform supervised fine-tuning on new capability-focused synthetic data mixed with..."
gpt-4o-2024-11-20	OpenAI	200	20000	100.1			6.7	95.7		46		Nov/2024		Material decrease in benchmark scores (GPQA: -13.37%, MMLU: -3.38%) compared to Aug/2024. Pruned? Quantized? https://github.com/openai/simple-evals
	DeepSeek-AI	67	2000	30.1			4.6			58.5		Nov/2024		o1 reasoning model copy. Scores 0/5 on latest ALPrompt 2024 H2 "DeepSeek-R1-Lite is currently still in the iterative development stage. It currently only supports web usage and does not support API calls. The base model used by DeepSeek-R1-Lite..."
Xmodel-LM	XiaoduoAI	1.1	2064	1,877.1			0.2	25.9				Nov/2024		SLM
	Pixtral Large	124	6000	49.1			2.9					Nov/2024		Open-weights multimodal model built on top of Mistral Large 2.
f1	Fireworks									42.4		Nov/2024		Compound
	Qwen2.5-Coder	32.5	5500	170.1			1.4	79.1				Nov/2024		"a compound AI model specialized in complex reasoning, that interweaves multiple open models at the inference layer."
Fox-1	TensorOpera	1.6	3005	1,879.1			0.2	44.99				Nov/2024		https://x.com/bloc97/status/1863675225810043331 "we introduce DisTrio, a family of architecture..."

	Lab	Parameters (B)	Tokens trained (B)	Ratio Tokens/Param ALScore = Tokens/Parameters ALScore is a quick and dirty rating of the model's power. The formula is: Sq Root of (Parameters * Tokens) * 100.	MLLM	MLLMU	GPQA	Training dataset	Announced	Public?	Y	Paper / Repo	Arch	Notes	
Qwen2-72B-A14B	Alibaba	57	4500	79.1	1.7	76.5	43	34.3	Jun/2024	Yes	Jun/2024	https://arxiv.org/abs/2407.18671	MoE	https://qwenlm.github.io/blog/qwen2/	
Skywork MoE 16x13B	Kunlun Tech	146				77.4			Jun/2024	Yes	Jun/2024	https://github.com/SkyworkML/Skywork-MoE/blob/main/skywork-moe-tech-report.pdf	MoE	CN + EN. (MoE) model with 146 billion parameters, 16 experts, and 22 billion activated parameters. This model is initialized from the pre-existing dense checkpoints of our Skywork-13B model."	
MAP-Neo	CMU	2.7	300	112.1		0.1			May/2024	Yes	May/2024	https://arxiv.org/abs/2405.23660	Dense	Analysis: https://ltdao.me/blog/2024/mamba2-part1-model/	
K2	International	7	4500	643.1		0.6	58.14		May/2024	Yes	May/2024	https://arxiv.org/abs/2405.19327	Dense	"first fully open-sourced bilingual LLM with comparable performance to existing state-of-the-art LLMs... we open-source all details to reproduce our MAP-Neo, where the cleaned pre-training corpus, data cleaning pipeline, checkpoints, and well-optim	
Codestral	LLM360	65	1400	22.1		1.0	64.8		May/2024	Yes	May/2024	https://www.lm360.ai/blog/overall-new-releases-by-further-our-mission.html	Dense	"K2-65B is a fully reproducible LLM outperforming Llama 2 70B using 35% less compute."	
Aya-23-35B	Mistral	22	2000	91.1		0.7			May/2024	Yes	May/2024	https://mistral.ai/news/codestral/	Dense	Fluent in 80+ programming languages	
Yi-Large	Cohere	35	4800	138.1		1.4			May/2024	Yes	May/2024	https://aihub.baai.ac.cn/files/1594860/01aiYi-Large-1.2ZiQinFeng788-Sovereign	Dense		
Yi-Large	01-ai	2000	20000	10.1		21.1	85.1	48.2	May/2024	Yes	May/2024	https://www.baai.ac.cn/news/01aiYi-Large-1.2ZiQinFeng788-Sovereign	MoE	Still training as of May/2024: https://appservearc.8bpc.cn/FnDYIEC4STBphu6M3NL4QKH43FEW dead link, use: https://finance.china.com.cn/tol/20240513/6116857.shtml	
Chameleon	01-ai	1000	15000	15.1		12.9	83.8	58.1	May/2024	Yes	May/2024	https://www.baai.ac.cn/news/01aiYi-Large-1.2ZiQinFeng788-Sovereign	Dense		
LearnLM	Meta AI	34	9200	271.1		1.9	65.8		May/2024	Yes	May/2024	https://arxiv.org/abs/2405.08818	Dense	Multimodal	
Sparse Llama 7B	Google DeepMind	1500	30000	20.1		22.4		72	May/2024	Yes	May/2024	https://arxiv.org/abs/2405.08818	MoE	Fine-tuned + prompted Gemini (Dec/2023). "The results of LearnLM-Tutor reproduce the performance of Gemini Pro, for example an MMLU score of 0.72 and MATH score of 0.33."	
1.5 Flash	Cerebras	7	145	21.1		0.1			May/2024	Yes	May/2024	https://arxiv.org/abs/2405.03954	Hybrid	https://www.cerebras.net/blog/introducing-sparse-llama-7b-smaller-3x-faster-full-accuracy "For the 50% sparse model, we utilized 45 billion tokens of pretraining data, while an additional 100 billion tokens were used for the 70% model. This represent	
GPT-4o	Google DeepMind					78.9	59.1	39.5	May/2024	Yes	May/2024	https://openai.com/index/gpt-4o-system-card	MoE	1M context length.	
GPT-4o	OpenAI	200	20000	100.1		6.7	88.7	72.6	53.9	May/2024	Yes	May/2024	https://openai.com/index/gpt-4o-system-card	MoE	Ominimodel. 1GPT-4o is likely an early checkpoint of GPT-5. https://twitter.com/dramfan/status/1790059671365757313 ELO: https://twitter.com/LiamFEdus/status/17900649639866370209 Demo: https://youtu.be/7QCaC89Daw
Fugaku-LLM	TII	11	5500	500.1		0.8	58.37		May/2024	Yes	May/2024	https://www.fujitsu.com/usa/news/technology-innovation-institute-releases-new-ai-model-series-outperforming-metals	Announce	Announce: https://www.llm.ai/news/falcon-2-uaes-technology-innovation-institute-releases-new-ai-model-series-outperforming-metals	
YOYO	Fujitsu	13	380	30.1		0.2			May/2024	Yes	May/2024	https://www.fujitsu.com/global/about/resources/news/press-releases/2024/05/01-01.html	Dense	Japanese CPU trained: 158.976+ A64FX CPUs (7M+ cores), zero GPUs https://en.wikipedia.org/wiki/Fugaku_Supercomputer	
YOYO	01-ai	34.4	3600	105.1		1.2	76.8	52.3	May/2024	Yes	May/2024	https://github.com/yiyi-1.5	Dense	Uses 600B more training tokens than Yi 1.0 (Nov/2023).	
YOYO	Microsoft	3	1600	534.1		0.2			May/2024	Yes	May/2024	https://arxiv.org/abs/2405.02454	Dense	With Tsingua. You Only Cache Once (YOCC). Long context "1M context length with near-perfect needle retrieval accuracy"	
DeepSeek-V2	DeepSeek-AI	236	6100	35.1		4.6	78.5	54.8	May/2024	Yes	May/2024	https://arxiv.org/abs/2405.02454	MoE	Huge dataset, 12% Chinese "Therefore, we acknowledge that DeepSeek-V2 still has a slight gap in basic English capabilities with LLaMA3 70B".	
ChuXin	Independent	1.6	2300	1,438.1		0.2	41.07		May/2024	Yes	May/2024	https://arxiv.org/abs/2405.02454	Dense	"results on the 'Needle In A Haystack' (NIAH) tests indicate that ChuXin-1M performs well across all context window lengths up to 1M."	
RWKV	RWKV	7.63	2500	328.1		0.5			May/2024	Yes	May/2024	https://github.com/ELLIS-LLM/Finch	Dense	Uses 600B more training tokens than Yi 1.0 (Nov/2023).	
xLSTM	ELLIS	2.7	15	6.1		0.0			May/2024	Yes	May/2024	https://arxiv.org/abs/2405.04517	Dense	New method LSTM to xLSTM, see also RNNs. Code/weights doesn't seem to be released. https://github.com/Al-Gurux/lstm-resources	
Granite Code	IBM	34	3500	103.1		1.1	50		May/2024	Yes	May/2024	https://github.com/IBM/granite/granite-code-models/blob/main/paper.pdf	Dense	MMLU=50 for 8B model only. Dataset: publicly available datasets (e.g., GitHub Code Clean, Starcoder data), public code repositories, and issues from GitHub.	
Qwen-Max	Alibaba	300	6000	20.1		4.5			May/2024	Yes	May/2024	https://arxiv.org/abs/2405.04517	Dense	https://twitter.com/justinliang10/status/1787584325367529509	
Med-Gemini-L 1.0	Google DeepMind	1500	30000	20.1		22.4			May/2024	Yes	May/2024	https://arxiv.org/abs/2404.18416	Dense	Med-Gemini-M 1.0 and Med-Gemini-L 1.0 (Pro and Ultra finetunes) "For language tasks that require less complex reasoning, such as summarizing medical notes and creating referral letters, we introduce Med-Gemini-M 1.0 by fine-tuning the Gemini	
TinyStories	Microsoft	0.033	50	1,516.1		0.0			Apr/2024	Yes	Apr/2024	https://arxiv.org/abs/2405.07739	Dense	Precursor to phi.	
Tele-FLM	BAAI	52	2000	39.1		1.1	64		Apr/2024	Yes	Apr/2024	https://arxiv.org/abs/2404.18646	Dense	Also known as FLM-2. "We will open-source a 1T model checkpoint, namely Tele-FLM-1T, to advance further training and research." Discussion paper Jul/2024: https://arxiv.org/abs/2407.02783	
Qwen-1.5 110B	Alibaba	111	3000	28.1		1.9	80.4	49.9	35.9	Apr/2024	Yes	Apr/2024	https://arxiv.org/abs/2404.18646	Dense	Worse performance on GPQA (72B=36.3, 110B=35.9).
Arctic	Snowflake AI Research	480	3500	8.1		4.3	67.3		Apr/2024	Yes	Apr/2024	https://arxiv.org/abs/2404.18646	Hybrid	"Arctic uses a unique Dense-MoE Hybrid transformer architecture. It combines a 10B dense transformer model with a residual 128+3.66B MoE MLP resulting in 480B total and 17B active parameters chosen using a top-2 gating."	
SenseNova 5.0	SenseTime	600	10000	17.1		8.2	84.78	42.93	Apr/2024	Yes	Apr/2024	https://arxiv.org/abs/2404.18646	MoE	GPT-4 scale; low media coverage; no demo in Western world. https://www.techinasia.com/sensetime-pauses-trading-stock-rises-30-model-launch	
OpenLM	Apple	3.04	1500	494.1		0.2	26.76		Apr/2024	Yes	Apr/2024	https://arxiv.org/abs/2404.18646	Dense	On-device model (laptop, phone). Open-source Efficient Language Models (OpenELM). https://venturebeat.com/ai/apple-releases-openelm-small-open-source-ai-models-designed-to-run-on-device/	
phi-3-medium	Microsoft	14	4800	343.1		0.9	78.2	55.7	Apr/2024	Yes	Apr/2024	https://arxiv.org/abs/2404.18646	Dense	Preview only, benchmarks being investigated as of May/2024.	
phi-3-mini	Microsoft	3.8	3300	869.1		0.4	68.8	45.7	Apr/2024	Yes	Apr/2024	https://arxiv.org/abs/2404.18646	Dense	"phi3-mini can be quantized to 4-bits so that it only occupies ~ 1.8GB of memory. We tested the quantized model by deploying phi-3-mini on iPhone 14 with A16 Bionic chip running natively on-device and fully offline achieving more than 12 tokens pe	
Llama 3 70B	Meta AI	70	15000	215.1		3.4	82	52.8	Apr/2024	Yes	Apr/2024				

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SOLAR-10.7B	Upstage AI	10.7								Dec/2023		https://arxiv.org/abs/2312.15196	Dense	South Korean. Llama-2 arch. SOTA for its size (Dec/2023).
DeciLM-7B	Deci	7.04	200	29.1		0.1				Dec/2023		https://arxiv.org/abs/231209034	Dense	4.4x times faster than Mistral. English only.
Mistral-medium	Mistral	180	3500	20.1		2.6	75.3			Dec/2023		https://mistral.ai/news/first-release/	Dense	MLLMU=75.3% (GPT-3.5-turbo 20B=70%, Llama 2 70B=68.9%)
mistral-8x7b-32kciqien	Together	46.7	8000	172.1		2.0	70.6	43.3		Dec/2023		https://www.together.ai/news/first-release	MoE	MoE=76.8x, aka mistral-small. Concretely, Mistral has 458B total parameters but only uses 12B parameters per token. It, therefore, processes input and generates output at the same speed and for the same cost as a 12B model.*
StripedHyena 7B	StripedHyena									Dec/2023		https://www.stripedhyena.ai/news/first-release	Dense	RedPajamas G41, new arch beyond just Transformers
NexusRaven-V2 13B	Nexusflowai	7.65								Dec/2023		https://github.com/nexusflowai/NexusRaven-V2/tree/master	Dense	Based on CodeLlama, 'surpasses GPT-4 by up to 7% in function calling success rates in human-generated use cases involving nested and composite functions.'
Gemini Ultra 1.0	Google DeepMind	1500	30000	20.1		22.4	83.7	35.7		Dec/2023		https://deepmind.google/pressroom/gemini-ultra-announcement/1_enxet.pdf	Dense	Original MMLU=83.7, MMLU=90.04 with prompting, Chinchilla (20.1), dense, maybe 600B-2000T.
Mamba	OMU	2.8	300	108.1						Dec/2023		https://arxiv.org/abs/2312.00292	Dense	The Pile, new arch beyond just Transformers. 2.7B MMLU=26.2, 7B MMLU=33.3.
LVM-3B	Berkeley/ULH	3	420	140.1		0.1	26.2			Dec/2023		https://arxiv.org/abs/2312.00785	Dense	Paper is 25MB. First Large Vision Model (LVM); no text. Based on Llama and LAION 5B (1.49B).
SeaLLM-13b	Alibaba	13	2000	154.1		0.5				Dec/2023		https://arxiv.org/abs/2312.00738	Dense	Llama 2 for Southeast Asian (SEA) languages: Vietnamese, Khmer, Malay, Lao, Tagalog, and Burmese
pglx-70b-online	Perplexity	70	2000	29.1		1.2				Nov/2023		https://blog.perplexity.ai/blog/introducing-pplx-online-llms	Dense	Web access. Higher 'freshness' and 'truth' scores.
SeamlessM4T-Large v2	Meta AI	2.3								Nov/2023		https://ai.meta.com/research/publications/seamless-multilingual-expressive-and-streaming-speech-translation/	Dense	Based on NLLB and older models. https://github.com/facebookresearch/seamless_communication
Q-Transformer	Google DeepMind									Nov/2023		https://deepmind.google/pressroom/q-transformer/	Dense	Robotics, builds on RT-1
Yuan 2.0	IEIT	102.6	288	3.1		0.6				Nov/2023		https://arxiv.org/abs/2311.15786	Dense	Chinese + EN dataset include The Pile: DM, arxiv, wikipedia, book3, stack exchange, Freetail and medical
MEDITRON	EPFL	70	2000	29.1		1.2				Nov/2023		https://arxiv.org/abs/2311.15079	Dense	Llama 2 trained on med data using NVIDIA Megatron-LM. 'outperforms Llama-2-70B, GPT-3.5 (text-davinci-003, 8-shot), and Flan-PaLM on multiple medical reasoning tasks.'
Transformers-Arithmetic	Microsoft	0.1	0.3	3.1		0.0				Nov/2023		https://arxiv.org/abs/2311.14737	Dense	Proving maths is not memorized. Uses GPT-2-style model. Sébastien Bubeck
Starling-7B	Berkeley	7	2000	286.1		0.4		37.9		Nov/2023		https://starling.cs.berkeley.edu/	Dense	Llama 2 7B -> OpenChat 7B -> Starling-7B (RLAIF)
Infection-2	Infection AI	1200	20000	17.1		16.3				Nov/2023		https://infection.ai/infection-2	Dense	'now the 2nd best LLM in the world'. Finished training 19/Nov/2023, waiting for fine-tuning and release.
Claude 2.1	Anthropic	130	2500	20.1		1.9	78.5			Nov/2023		https://www.anthropic.com/index/claude-2-1	Dense	Less hallucinations, 200k context length, tool use
TÖLU 2	Allen AI	70	2000	29.1		1.2				Nov/2023		https://arxiv.org/abs/2311.15702	Dense	Llama 2 finetune with RLHF direct preference optimization (DPO).
Nemotron-3 22B	NVIDIA	22	3800	173.1		1.0	54.4			Nov/2023		https://developer.nvidia.com/blog/nvidia-ai-foundation-models-build-custom-enterprise-chatbots-and-co-pilots-with-production-ready-llms/	Dense	8B released, 22B internal.
Nemotron-2 43B	NVIDIA	43	3800	89.1		1.3				Nov/2023		https://arxiv.org/abs/2311.09528	Dense	Used to train HelpSteer (16/Nov/2023): https://arxiv.org/abs/2311.09528
Orca 2	Microsoft	13	2001	154.1		0.5				Nov/2023		https://arxiv.org/abs/2311.11045	Dense	Llama 2 13B (2T) -> Orca 2 (GPT-4 finetune). Still an imitation model, overhyped: The False Promise of Imitating Proprietary LLMs https://arxiv.org/abs/2305.15717
Phi-2	Microsoft	2.7	1400	519.1		0.2				Nov/2023		https://twitter.com/SebastianBubeck/status/1724854157004190095	Dense	https://twitter.com/SebastianBubeck/status/1724854157004190095
Florence-2	Microsoft	0.771								Nov/2023		https://arxiv.org/abs/2311.06242	Dense	VLM, Flamingo alt
Miraso3B	Google DeepMind	3								Nov/2023		https://arxiv.org/abs/2311.06080	Dense	Combiner + autoregressive transformer for video/audio/text
OtterHD-8B	NTU	8	737	93.1		0.3				Nov/2023		https://arxiv.org/abs/2311.04419	Dense	Evolution of PerSimmon-9.3B and Fuyu 8B
Gauss	Samsung	7								Nov/2023		https://samsungai.github.io/llm/gauss/	Dense	Gauss Language specializing in generating texts, Gauss Code on software and code description and Gauss Image for image creation.
Grok-1	xAI	314	6000	20.1		4.6				Nov/2023		https://github.com/xai-org/grok-1	Dense	Context window=8192. UI: https://twitter.com/TobyPhin/status/1721053802236621734
Grok-0	xAI	33	2000	61.1		0.9				Nov/2023		https://web.archive.org/web/20221106051542/https://x.ai/	Dense	Announced Nov/2023, trained Jul/2023
Yi-34B	01.ai	34.4	3000	88.1		1.1	76.3	43		Nov/2023		https://github.com/01-ai/Yi	Dense	Controversy about Llama 2 base. https://twitter.com/kailifee/status/1724673131875377465 (PaLM 2=78.3) Outperforms Llama 2. Chinese and English. https://www.bloomberg.com/news/articles/2023-11-05/kai-fu-lee-s-open-source-01-ai
GPT-4 Turbo	OpenAI		13000			86.4		46.5		Nov/2023		https://openai.com/blog/new-models-and-developer-products-announced-at-devday	MoE	https://openai.com/blog/new-models-and-developer-products-announced-at-devday
Skywork-13B	Kunlun Tech	13	3200	247.1		0.7	62.7			Oct/2023		https://arxiv.org/abs/2310.18341	Dense	CN + EN.
Kimi Chat	Moonshot AI	100	2000	20.1		1.5				Oct/2023		https://www.chinadaily.com.cn/a/20240322/WS65f8ed78a1108c04a1ee11.html	Dense	Chinese. Long context. No paper.
jina-embeddings-v2	Jina AI	0.435								Oct/2023		https://jina.ai/news/jina-ai-launches-worlds-first-open-source-8k-text-embedding-creating-openai/	Dense	Alternative to text-embedding-ada-002. Related v1 paper: https://arxiv.org/abs/2307.11224
Fuyu	Adept	8								Oct/2023		https://www adept.ai/blog/fuyu-8b	Dense	VLM. 8B available under open licence, Medium size is closed
ERNIE 4.0	Baidu	1000	20000	20.1		14.9				Oct/2023		https://baidu.com/technology/ernie-4-0-ai-model-unleash-erotic-ai-erme-ai-model-2023-10-11	Dense	Dense (confirmed), English-dubbed launch video (2h52m): https://twitter.com/vbroadcasts/1yNgaZaeelJJ & https://youtu.be/wYozcsavRUM
Zephyr	HuggingFace	7.3	800	110.1		0.3		33		Oct/2023		https://huggingface.co/luping/facelift/zephyr-7b-alpha	Dense	Mistral with 'aligned' data removed from dataset
PaLI-3	Google DeepMind	5								Oct/2023		https://arxiv.org/abs/2310.08189	Dense	VLM. Next iteration of PaLI via Pathways. https://tfresearch.at/pathways/
Retro 48B	NVIDIA	48	1200	25.1		0.8				Oct/2023		https://arxiv.org/abs/2310.07713	Dense	the largest LLM pretrained with retrieval before instruction tuning.'
Ferret	Apple	13	2000	154.1		0.5				Oct/2023		https://arxiv.org/abs/2310.07704	Dense	Vicuna base, multimodal
Lemur	XLANG Lab	70	2090	30.1		1.3				Oct/2023		https://arxiv.org/abs/2310.06830	Dense	
AceGPT	KAUST/Shenzhen	13	2010	155.1		0.5				Oct/2023		https://github.com/FreedomIntelligence/AceGPT/tree/main	Dense	Arabic. Llama 2 + RLAIF
Yasa-1	Reka AI									Oct/2023		https://reka.ai/product	Dense	Multi-modal. No public arch info. Researchers from DeepMind, Google, Baidu and Meta building enterprise models
RT-X	Google DeepMind	55								Oct/2023		https://robotics-transformer.github.io/paper.pdf	Dense	Robotics using UL2. RT-1 model trained using the robotic data mixture as RT-1-X, and the RT-2 model trained using the robotic data mixture as RT-2-X.*
Waymo	Motion	0.09								Sep/2023		https://arxiv.org/abs/2309.14524	Dense	LLM for autonomous vehicle forecasting. https://youtu.be/1M3z1182e1560
GALA-1	Wayve									Sep/2023		https://arxiv.org/abs/2309.12080	Dense	World model, generates video. Uses T5-large 770M for language + all vision parameters
Owen	Alibaba	72	3000	42.1		1.5				Sep/2023		https://arxiv.org/abs/2309.15609	Dense	Chinese. Full name is 'Tongyi Qianwen' 通义千问. 'Lags behind both GPT-3.5 and GPT-4'. Originally 7B/14B params Apr/2023
Llama 2 Long	Meta AI	70	2400	35.1		1.4				Sep/2023		https://arxiv.org/abs/2309.18039	Dense	Unreleased to date. Context window=32.768 tokens (compare to Llama 2=4096 tokens)
LeoLM	Hessian AI/LAION	13	2065	159.1		0.5				Sep/2023		https://arxiv.org/abs/2309.18039	Dense	Llama 2 'extended' and pretrained on 2000B Llama 2 tokens + 65B tokens of German
Mistral 7B	Mistral	7.3	800	110.1		0.3		30.9		Sep/2023		https://mistral.ai/news/first-release/	Dense	Apache 2.0, Sliding Window Attention (SWA) to handle longer sequences at smaller cost
Kosmos-2.5	Microsoft	1.3								Sep/2023		https://arxiv.org/abs/2309.11418	Dense	Great paper. Chinese-English bilingual dataset
Baichuan 2	Baichuan	13	2600	200.1		0.6				Sep/2023		https://data.baichuan.ai/composer/Baichuan2-technical-report.pdf	Dense	CPU trained
BOLT2.5B	ThirdAI	2.5	40	16.1		0.0				Sep/2023		https://medium.com/thirdai/blog/creating-the-worlds-first-generative-llm-are-trained-only-on-cpus-most-therpse-a-bolt2-5b-10c900e1c4d4	Dense	Faster inference (4.8x throughput of Llama 2)
DeciLM	Deci	5.7	200	36.1		0.1				Sep/2023		https://arxiv.org/abs/2309.04540	Dense	ModuleFormer is based on the Sparse Mixture of Experts (MoE).
MoLM	IBM	8	300	38.1		0.2				Sep/2023		https://arxiv.org/abs/2309.04540	MoE	ModuleFormer is based on the Sparse Mixture of Experts (MoE).
NEXT-GPT	Singapore	7	1000	143.1		0.3				Sep/2023		https://arxiv.org/abs/2309.04540	Dense	Multimodal. Vicuna 7B + other modalities
Phi-1.5	Microsoft	1.3	150	116.1		0.0				Sep/2023		https://arxiv.org/abs/2309.04540	Dense	Textbooks only. 30B-token dataset
UnILM	Apple	0.034	1	30.1		0.0				Sep/2023		https://github.com/apple/UnILM	Dense	Apple's Transformer model for iOS 17 + macOS Sonoma. Announcement is actually Jun/2023. GPT-2 base? 128 token context window
PerSimmon-8B	Adept	8	737	93.1		0.3				Sep/2023		https://github.com/persimmon-ai/labelled-inference	Dense	Open Apache license and publicly accessible weights.
FLM-101B	BAAI	101	245	3.1		0.5				Sep/2023		https://arxiv.org/abs/2309.03852	Dense	Train for 100k compute budget (on a cluster of 24 DGX-A800 GPU 8x80G servers for 21 days)
Falcon 180B	TII	180	3500	20.1		2.6	70.6			Sep/2023		https://arxiv.org/abs/2311.16807	Dense	Major milestone for open source models (largest open dense model to date).
Hunyuan	Tencent	100	2000	20.1		1.5				Sep/2023		https://arxiv.org/abs/2402.01723v1	Dense	
phi-CTRL	Independent	0.1	0.01	1.1		0.0				Sep/2023		https://arxiv.org/abs/2309.08832	Dense	Satire. MMLU=100, 'phi-CTRL (pronounced "fictional") that achieves perfect results across diverse academic benchmarks'
Granite	IBM	13	2500	193.1		0.6	57			Sep/2023		https://www.ibm.com/downloads/cas09W4C06M	Dense	Original training on 1T tokens, update 15Feb/2024 trained on 2.5T tokens: granite-13b-chat-v2 (v2.1.0). *AI IBM, we curated 6.48TB of data to train our LLM Granite.13B. This was reduced to 2.07 TB after pre-processing, a 68% decrease.*
Jais	Inception	13	395	31.1		0.2				Aug/2023		https://arxiv.org/abs/2308.16149	Dense	Arabic, trained in Abu Dhabi. UAE using Cerebras.
Code Llama 34B	Meta AI	34	2600	77.1		1.0				Aug/2023		https://ai.meta.com/research/publications/code-llama-open-foundation-models-for-code/	Dense	Outperforms GPT-3.5. Initial Llama 2 (2T tokens) trained on 500B tokens of code, 100B tokens of python
IDEFICS	HuggingFace	80								Aug/2023		https://huggingface.co/huggingface/idefics	Dense	Clone of Flamingo using Llama-1 65B. Named after Asterix and Obelix's dog Idexif (Image-aware Decoder Enhanced à la Flamingo with Interleaved Cross-attentionS)
Raven	UI/NVIDIA	11	40	4.1		0.1				Aug/2023		https://arxiv.org/abs/2308.07292	Dense	ROG Atlas
DukunLM	Azalea AI	13	1500	116.1		0.5				Aug/2023		https://huggingface.co/azalea-ai/DukunLM-13B-V1.0-Llama2-7B	Dense	Indonesian fine-tune of WizardLM (which is a Llama fine-tune).
WizardLM	Microsoft	70	2000	29.1		1.2				Aug/2023		https://github.com/microsoft/Wizard		

Model	Lab	Parameters (B)	Tokens trained (B)	Ratio Tokens/Param ALScore (Chinchilla scaling@20:1)	ALScore* (The formula is: Sqr Root of (Parameters x Tokens) + 300)	MLLM	MMLU	GPQA	Training dataset	Announced	Public?	Y	Paper / Repo	Arch	Notes
Jurassic-2	AI21	178								Mar/2023			https://www.ai21.com/blog/introducing-2	Dense	
OPT-NeoX-Chat-Base-20B	Together	20						33.6		Mar/2023			https://github.com/together-inference/OpenChat2	Dense	
Kosmos-1	Microsoft	1.6	360	225:1				0.1		Feb/2023			https://github.com/microsoft/Kosmos-1	Dense	Instruction-tuned 20 billion parameter language model, a 6 billion parameter moderation model, and an extensible retrieval system for including up-to-date responses from custom repositories. It was trained on the OIG-43M training dataset, which was
LLaMA-65B	Meta AI	65	1400	22:1				1.0	66.9	Feb/2023			https://arxiv.org/abs/2302.14655	Dense	Proto-AGI. Multimodal large language model (MLLM). Raven's Progressive Matrices as real images, not digits as in testing of text-davinci-003 at https://lmsys.org/demos/llm-arena/
MOSS	Fudan University	16	430	27:1				0.3		Feb/2023			https://arxiv.org/abs/2302.14655	Dense	Researchers only, noncommercial only. LLaMA-65B is competitive with the best models, Chinchilla70B and PaLM-540B.
Palmyra	Writer	20	300	15:1				0.3		Feb/2023			https://arxiv.org/abs/2302.14655	Dense	Major bandwidth issues: https://www.reuters.com/technology/china-fudan-university-team-apologizes-after-chatgpt-style-platform-crashes-2023-02-21/
Luminous Supreme Control	Aleph Alpha	70	588	9:1				0.7		Feb/2023			https://www.aleph-alpha.com/pdf/2023_02_AA_Benchmarks_doc.pdf	Dense	Only up to 5B available open-source 'trained on over 300 billion tokens of text data, and the size of the resulting model is over 20 billion parameters.' https://writer.com/product/cowriter/
Toolformer+Atlas 11B+NLB 54B	Meta AI	6.7	402	60:1				0.2		Feb/2023			https://arxiv.org/abs/2302.04761	Dense	'Control' means instruction tuned
Multimodal-CoT	Amazon	0.738								Feb/2023			https://arxiv.org/abs/2302.03923	Dense	Based on GPT-J 6.7B + access to other models via API
FLAME	Microsoft	0.06	9	150:1				0.0		Jan/2023			https://arxiv.org/abs/2301.12779	Dense	Models <1B with vision CoT
Med-PaLM 1	Google DeepMind	540	780	2:1				2.2		Dec/2022			https://arxiv.org/abs/2212.13138	Dense	T5 for Excel formulas, very small 60M params, "We start from a dataset of 927M formulas" estimate 10x multiplier for 9B tokens
OPT-IML	Meta AI	175	300	2:1				0.8		Dec/2022			https://arxiv.org/abs/2212.12017	Dense	Collab between Google & DeepMind. Makes 1% less errors than humans
RL-CAI	Anthropic	52	400	8:1				0.5		Dec/2022			https://arxiv.org/abs/2212.08073	Dense	Instruct
ERNIE-Code	Baidu	0.56								Dec/2022			https://arxiv.org/abs/2212.06749v2	Dense	RLAI=reinforcement learning with AI feedback
RT-1	Google	0.035								Dec/2022			https://robotics-transformer.github.io/assets/rt1.pdf	Dense	
ChatGPT (gpt-3.5-turbo)	OpenAI	20						70	28.1	Nov/2022			https://openai.com/blog/chatgpt	Dense	Instruct with strict policies ("extremely limited")
text-davinci-003	OpenAI									Nov/2022			https://openai.com/blog/chatgpt	Dense	
GPT-JT	Together	6								Nov/2022			https://www.together.ai/blog/introducing-v1-of-gpt-j-powered-by-open-source-ai	Dense	
RWKV-4	RWKV	14	332	24:1				0.2		Nov/2022			https://arxiv.org/abs/2209.12648	Dense	RNN, not transformer: https://www.reddit.com/r/MachineLearning/comments/vxt8sar/v1_of_gpt_j_release_an_attentionfree_nn_language/
Galactica	Meta AI	120	450	4:1				0.8	52.6	Nov/2022			https://galactica.org/whitepaper.pdf	Dense	scientific only
SED	DeepMind									Nov/2022			https://arxiv.org/abs/2211.04226	Dense	SED 420M (diffusion text model)
mT0	BigScience	13	1000	77:1				0.4		Nov/2022			https://arxiv.org/abs/2211.01786	Dense	fine-tuned
BLOOMZ	BigScience	176	366	3:1				0.8		Nov/2022			https://arxiv.org/abs/2211.01786	Dense	fine-tuned
PACT	Microsoft		0.03					0.0		Oct/2022			https://arxiv.org/abs/2208.11133	Dense	Trained on ~5TB data, 2GB model download. 'In general we see an improvement in model performance as we increase the number of training tokens. Interestingly, larger models did not necessarily result in better performance for robot navigation. Ev
Flan-T5	Google	11	1100	100:1				0.4		Oct/2022			https://arxiv.org/abs/2208.12438	Dense	T5=1T tokens + LM-adapted T5 as 100B tokens
Flan-PaLM	Google	540	780	2:1				2.2	73.5	Oct/2022			https://arxiv.org/abs/2210.11418	Dense	
U-PaLM	Google	540	780	2:1				2.2	74.1	Oct/2022			https://arxiv.org/abs/2210.11499	Dense	
NVIDIA	NVIDIA	0.2								Oct/2022			https://arxiv.org/abs/2210.00904	Dense	
VIMA	OpenChat	13	2000	154:1				0.5		Sep/2022			https://arxiv.org/abs/2209.11298	Dense	Llama 2 13B -> OpenChat 13B
Taighua	Wechat	10	300	66:1				0.2		Sep/2022			https://arxiv.org/abs/2209.10372	Dense	13% English tokens and 87% Chinese
CodeGeex	Taighua	13	850	30:1				0.4		Sep/2022			https://github.com/THUDM/CodeGeex	Dense	
DeepMind	Google	70	1400	20:1				1.0		Sep/2022			https://storage.googleapis.com/deepmind-media/DeepMind.com/authors/index/openai/gpt3e.html	Dense	Chatbot as a fine-tuned version of Chinchilla 70B
PaLI	Google	17								Sep/2022			https://arxiv.org/abs/2209.06794	Dense	PaLM Vision model, new datasets of 10B multilingual text-image pairs
NeMo Megatron-GPT 20B	NVIDIA	20								Sep/2022			https://huggingface.co/nvidia/nemo-megatron-gpt-20b	Dense	
Z-Code++	Microsoft	0.71	500	705:1				0.1		Aug/2022			https://arxiv.org/abs/2208.08770v1	Dense	abstractive text summarization, 710M, outperforms PaLM 540B. "Due to the limited computational resource, Z-Code++LARGE is trained with only 500B tokens instead of 1T tokens as that for mT5 training."
Atlas	Meta AI	11	40	4:1				0.1	47.9	Aug/2022			https://arxiv.org/abs/2208.02269	Dense	
BlenderBot 3	Meta AI	175	300	2:1				0.8		Aug/2022			https://github.com/facebookresearch/blenderbot/main/projects/bl3/bl3_main_tech_report.pdf	Dense	
GLM-130B	Taighua	130	400	4:1				0.8		Aug/2022			https://arxiv.org/abs/2210.02414	Dense	50% English (200B tokens), so included here
AlexaTM 20B	Amazon	20	1300	65:1				0.5		Aug/2022			https://research.amazon.science/2023/02/30/alexa20b/alexa20b17/alexa20b-fine-tuned-learning-using-a-large-scale-multilingual-and-qa-model.pdf	Dense	Wikipedia and mC4 only. seq2seq
6.9B FIM	OpenAI	6.9	100	15:1				0.1		Jul/2022			https://arxiv.org/pdf/2207.14255.pdf	Dense	Several models: 8 sizes, NLP, Code, FIM/non-FIM. 100B tokens for 6.9B params... beyond chinchilla
'monorepo-Transformer'	Google	0.5								Jul/2022			https://ai.googleblog.com/2022/07/monorepo-code-completion-improves.html	Dense	Unnamed. Writes >3% of internal google code
PanGu-Coder	Huawei	2.6								Jul/2022			https://arxiv.org/abs/2207.11280	Dense	Python via GH
NLLB	Meta AI	54.5								Jul/2022			https://research.facebook.com/publications/nllb-language-left-behind/	MoE	54.5B MOE, 3.3B dense. 200+ languages
J-1 RBG	AI21	178	300	2:1				0.8		Jul/2022			https://www.ai21.com/blog/announcing-ai21-studio-and-jurassic-1	Dense	J-1 fine-tuned with RBG law corpus
BLOOM (tr11-176B-ml)	BigScience	176	366	3:1				0.8	39.1	Jul/2022			https://github.com/bigscience-workshop/bigscience/tree/master/chains/11-176B-ml	Dense	
Minerva	Google	540	818.5	2:1				2.2		Jun/2022			https://ai.googleblog.com/2022/06/minerva-solving-quantitative-reasoning.html	Dense	PaLM finetuned on LaTeX/arXiv maths
GODEL-XL	Microsoft	2.7								Jun/2022			https://arxiv.org/abs/2208.11309v1	Dense	XL: GPT-3 175B in paper, GPT-J 2.7B released
YaLM 100B	Yandex	100	300	3:1				0.6		Jun/2022			https://github.com/yandex/yaLM-100B	Dense	Megatron-LM clone, Russian/English: https://medium.com/yandex/yandex-publishes-ya1m-100b-its-the-largest-gpt-like-neural-network-in-open-source-d1df53d0e9a6
Unified-IO	Allen AI	2.8								Jun/2022			https://github.com/allenai/unified-speech2text/unifiedio1.pdf	Dense	Based on T5. Demo only
Perceiver AR	DeepMind	1								Jun/2022			https://arxiv.org/abs/2208.07769	Dense	Context window=100,000. Params=364m wkt, 975M pg-19, 826M books, music=?, imagenet=770M.
LIME	Google	5.6								Jun/2022			https://ai.googleblog.com/2022/06/limo-fine-tuning-multitask-models-with-hpa	MoE	
GPT-4chan	Independent	6								Jun/2022			https://arxiv.org/abs/2001.07487	Dense	Warning for inappropriate content, GPT-J.
Diffusion-LM	Stanford	0.3								May/2022			https://arxiv.org/abs/2208.14217	Dense	GPT-J with synthetic data
UL2 20B	Google	20	1000	50:1				0.5	39.2	May/2022			https://arxiv.org/abs/2205.03131	Dense	Unifying Language model. C4 only.
Gato (Cat)	DeepMind	1								May/2022			https://storage.googleapis.com/deepmind-media/Gato/Generators%20Agent.pdf	Dense	Proto-AGI. Generalist agent (LLM, VLM, robot)
LaMDA 2	Google	137								May/2022			https://arxiv.org/abs/2201.08239	Dense	Chatbot with tiny walled garden demo TBA
OPT-175B	Meta AI	175	300	2:1				0.8		May/2022			https://arxiv.org/abs/2204.07709	Dense	Only 30B available (Jun/2022)
TK-Instruct	Hugging Face	11								Apr/2022			https://arxiv.org/abs/2204.07709	Dense	Based on T5.
Meta AI	InCoder	6.7								Apr/2022			https://arxiv.org/abs/2204.06999	Dense	Python and JavaScript
NOOR	TII	10								Apr/2022			https://www.tii.ac/research/technology-innovation-institute-announces-launch-noor-writs-largest-arabic-nlp-model	Dense	Arabic. "World's largest high-quality cross-domain Arabic dataset, combining web data with books, poetry, news articles, and technical information"
mOPT	Sber	13								Apr/2022			https://arxiv.org/abs/2204.07280	Dense	60 languages. Only 1.3B model available
PaLM-Coder	Google	540	780	2:1				2.2		Apr/2022			https://storage.googleapis.com/pathways/language-model/PaLM-paper.pdf	Dense	
PaLM	Google	540	780	2:1				2.2		Apr/2022			https://storage.googleapis.com/pathways/language-model/PaLM-paper.pdf	Dense	
SeeKeR	Meta AI	2.7								Mar/2022			https://arxiv.org/abs/2203.13224	Dense	BART and compared to GPT-2
CodeGen	Salesforce	16								Mar/2022			https://arxiv.org/abs/2203.13474	Dense	Code
VLM-4	LightOn	10								Mar/2022					